Automating Variant Drug Annotations with Schema Based Constraints

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Introduction

Pharmacogenomics aims to personalize medicine by optimizing drug therapies based on genetic profiles, but its clinical integration is hindered by the extensive manual effort required to annotate the vast number of genetic variants in biomedical literature. Previous efforts, such as PGXMine, have introduced automation to this process; however, success is limited by by local context understanding. Large language models (LLMs) offer significant advantages for natural language understanding and capture nuanced context to perform complex information extraction tasks. However, their outputs can be unpredictable and inconsistent, particularly when dealing with highly complex data like pharmacogenomic annotations. A schema, which provides a structured framework defining relationships and constraints among data elements, can address this unpredictability by guiding the model's outputs to align with predefined data formats and rules.
This study evaluates schema-constrained GPT-4 for automating variant-drug data extraction, leveraging the contextual strength of LLMs while mitigating their limitations through schema constraints.

Methods

We use the publicly available Pharmacogenomics Knowledge Base(PharmGKB) variant-drug annotations as a ground truth and reference for articles. We select 5000 samples from the variant annotation dataset and retrieve the content of annotation article autonomously through playwright and bioentrez python packages. 656 full articles were scraped. For each of these articles we convert the webpage Hyper Text Markup Language(HTML) to text with delimited spacing, and provide as context to language model such as GPT4o. We force Java Script Object Notation(JSON) schema on the model to return the following five categories "gene", "drug(s)", "phenotype category", "significance", "metabolizer types" with hard constraints on phenotype category and metabolizer types. Model outputs were matched by Gene and PMID, then either exactly matched per aligned variant row, or grouped by PMID and matched by intersection of sets.

Fig. 1 Preprocessing Overview

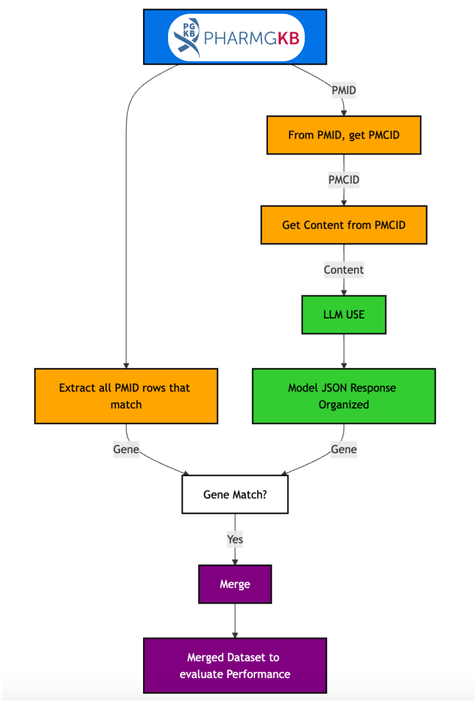


Fig. 2 Javascript Object Notation Schema and example

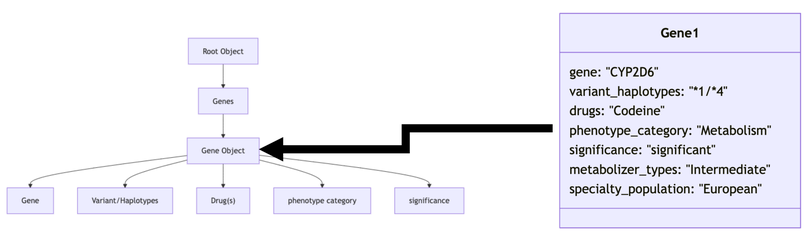
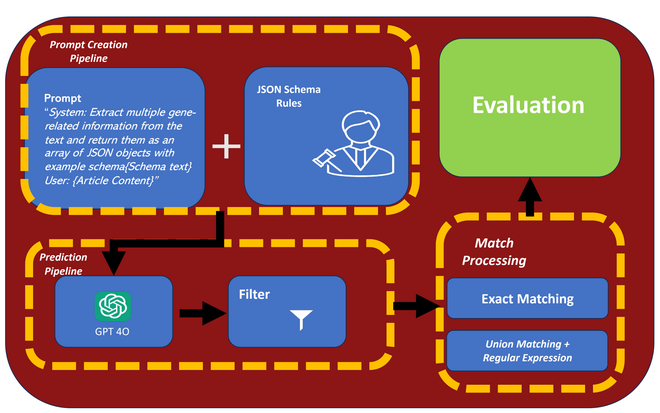


Fig. 3 Methods Pipeline Overview



Results

Fig. 4 Example Outputs

Variant/Haplotypes Truth	Gene Truth	Drug(s)_truth	Phenotype Category Truth	Significance	Variant/Haplotypes Predicted	Gene Predicted	Drug(s) Predicted	Phenotype Category Predicted	significance
rs2952768	METTL21A	morphine	Dosage	yes	rs1799971 (A>G)	OPRM1	morphine	Metabolism/PK	significant
rs1799971	OPRM1	morphine	Dosage	yes	rs1799971 (A>G)	OPRM1	morphine	Metabolism/PK	significant
rs2228570	VDR	amoxicillin, clarithromycin, omeprazole	Efficacy	yes	rs2228570, rs7975232	VDR	clarithromycin, amoxicillin, omeprazole	Efficacy	significant
rs7975232	VDR	amoxicillin, clarithromycin, omeprazole	Efficacy	yes	rs2228570, rs7975232	VDR	clarithromycin, amoxicillin, omeprazole	Efficacy	significant
rs2228570	VDR	amoxicillin, clarithromycin, omeprazole	Efficacy	yes	FokI, ApaI	VDR	clarithromycin, amoxicillin, omeprazole	Efficacy	significant

Fig. 5 Exact vs Any Match

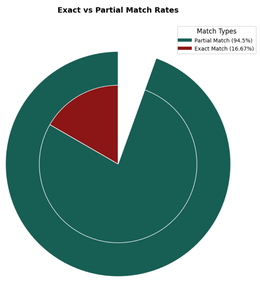
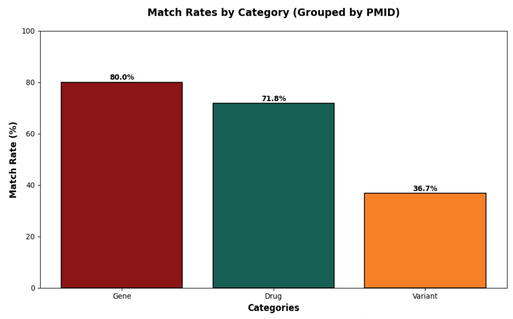
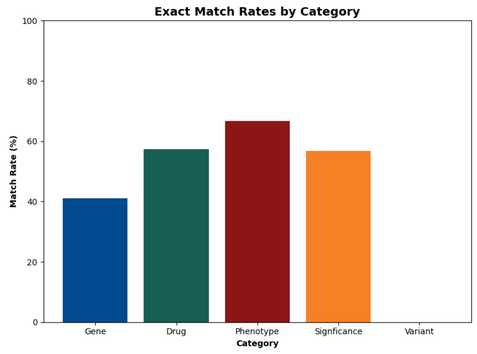


Fig. 6 Grouped Match



***For missing categories exact matches are not differentiated well.*

Fig. 7 Exact Match



Conclusions

Overall, even with constraints, GPT4O struggles to keep precision but still performs moderally well for, **Gene**(41%), **Drugs**(57.4%.), **Phenotype**(66.68%), and **Significance**(56.75%), but for exact **Variants**(0%) fail.

Grouping by PMID, GPT-4O improves for **Gene**(79.99%), **Variants**(36%), and **Phenotype**(71.18%).

While significance and other categories can be observed with grouping by PMID, it would not make sense when variant relationships are independent of each other.

While schema-constrained GPT-4 demonstrates promise in reducing the manual workload of pharmacogenomic annotation, it is limited in accuracy and alignment.

These findings highlight both the opportunities and pitfalls of using large language models to advance pharmacogenomics, emphasizing the need for careful

Future Steps and Limitations

While we attempted to mitigate information loss from tables by leveraging HTML, our efforts were constrained by the lack of table-specific awareness in the model. This limitation often resulted in the omission of crucial information essential for variant identification and annotation.

For variants, discrepancies between star alleles and rsIDs posed additional challenges, contributing to potential errors. This issue was evident when outputs occasionally included exact Single Nucleotide Polymorphisms (SNPs) without adequate contextual alignment. Furthermore, in instances where information was missing, the model frequently generated confabulated results, even when explicitly permitted to leave outputs empty.

Language models show significant promise as tools for augmenting or automating annotation workflows but require external support to achieve robust and accurate results.
To address these challenges, we aim to explore agent-based systems to enhance and streamline this workflow.

References

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